Arpan Pathak

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Visa: Requires H1B Transfer sponsorship

SUMMARY

Systems Software Engineer with 8+ years architecting high-performance, developer-facing SDKs, APIs, and distributed systems at **Amazon**, **Microsoft**, and **Oracle**. Deep expertise in **C++**, **Python**, **GPU-accelerated computing**, and **performance-critical software engineering —** building low-latency, high-throughput engines at global scale. Proven track record delivering production-grade software with rigorous CI/CD, testing/validation, containerization, and Kubernetes infrastructure. Passionate about **computer graphics**, **neural reconstruction**, and **Physical AI —** actively exploring **CUDA** and **Rust-to-CUDA (CUDA-Oxide)** paradigms to bridge real-time rendering and simulation.

PERSONAL PROJECTS

Driving-CivicSense-Vision-Model | github.com/arpanpathak/driving-civicsense-vision-model | 2026

- AI-driven auxiliary perception system built in **Rust** for real-time intersection discipline and lane-awareness — runs entirely on-device (Edge AI, < 50ms latency, no cloud dependency).
- Combines **YOLOv8/v11** computer vision models with custom inference pipeline to detect stop sign compliance, intersection occupancy violations, and left-lane camping.
- Trained on cloud GPU infrastructure with focus on **real-time performance**, privacy-first architecture, and aftermarket dashcam/AR glasses deployment.

EXPERIENCE

Software Developer II Microsoft — Redmond, WA | June 2025 – Present

- **C++ to Rust Migration**: Spearheaded the migration of the Information Protection service from **C++ to Rust**, shifting memory safety to compile-time. Eliminated use-after-free and buffer-overflow CVEs in the audited path while maintaining full performance parity — a critical win for DORA compliance.
- **AI-Powered Policy Engine**: Built an AI-driven engine that auto-generates network policies and scans for misconfigurations, reducing manual policy review cycles from 4 hours to near-real-time with zero policy drift across sovereign regions.
- **eBPF & Kernel-Level Networking**: Designed and deployed eBPF-based network policies using **Cilium** for Microsoft Purview's sovereign cloud regions. Authored custom **Kubernetes Operators** (Kubebuilder) to reconcile compliance states across AKS clusters.
- **Developer Workflows**: Established CI/CD pipelines, containerized microservices, and automated validation frameworks ensuring reliable multi-region releases.
- Infra: **C++, Rust, Python, Kubernetes (AKS), Kubebuilder, Docker, CI/CD, Linux**

Senior Software Engineer Oracle Cloud Infrastructure — Seattle, WA | Oct 2024 – Mar 2025

- **High-Performance Engine**: Architected low-latency internals for a proprietary database engine, optimizing **lock-free data structures** to reduce contention in high-concurrency read-write paths, improving transaction throughput by **45,000 ops/s**.
- **Infrastructure-as-Code SDK**: Engineered Terraform provider (developer-facing SDK) for Oracle Autonomous Database, replacing legacy control plane infrastructure with projected **$1.2M/month cost savings**.
- **Security & Key Management**: Integrated OCI Security Vault across control plane services for secure secret management and automated key rotation protocols.
- Infra: **C++, Rust, Go, Terraform, OCI, PostgreSQL internals**

Software Development Engineer II Amazon — Seattle, WA & Hyderabad, India | Oct 2021 – Oct 2024

- **Real-Time ML Ranking Engine**: Architected an **ensemble model pipeline combining XGBoost and Deep Learning** for real-time ads & deal ranking across **FreeVee, Twitch, MiniTV, Prime Video, and Amazon Retail**, driving **5% CTR increase**, 30% higher deal impressions, and **$5.4M monthly revenue growth** — processing millions of events with **100ms latency**.
- **SDK & Distributed Systems Design**: Designed a uniqueness constraint indexing solution and **SDK** for a multi-tenant NoSQL datastore using two-phase commit, achieving **serializable isolation** at 3K writes/second — a developer-facing library used across multiple publisher teams.

- **Conversational AI (BERT)**: Developed a **BERT-based AI model** handling **5,000 QPS** at peak, reducing support ticket triage SLA from 7 days to 2 hours.
- **CI/CD & Developer Productivity**: Led migration to React.js microfrontend architecture with automated CI/CD pipelines, reducing deployment cycles from **5 days to 30 minutes —** enabling full team autonomy and faster iteration.
- **Big Data Pipelines**: Engineered **Apache Spark** jobs processing millions of daily events into Amazon Redshift for OLAP with automated dashboards.
- Infra: **Python, C++, Java, Spark, Kafka, DynamoDB, Redshift, SageMaker, AWS, Docker**

Software Development Engineer Razorpay — Bengaluru, India | Aug 2020 – Feb 2021

- Built UPI Payment Gateway Aggregator using **Golang** and Protobuf, processing **1M+ hourly transactions**.
- Maintained **99.99% uptime** with **150ms P99 latency** under 50,000 TPM peak loads.
- Infra: **Go, Kafka, Redis, PostgreSQL**

Software Engineer Mindfire Solutions — Bhubaneswar, India | Aug 2018 – Feb 2020

- Developed full-stack workflow tools and **AR-based product search microservice**, serving 10,000+ daily queries.
- Built administrative dashboards reducing data retrieval latency from 5 min to 1 min.
- Infra: **Java, React, Python, MongoDB**

SKILLS

Languages: Python, C++, Rust, Go, Java, Kotlin, TypeScript, C#, Shell Script

GPU & Graphics: CUDA (exploring), CUDA-Oxide, GPU-Accelerated Computing, Real-Time Rendering, Computer Graphics

SDK & Developer Tools: SDK Design, API Development, Developer-Facing Libraries, Terraform Providers, CI/CD, Automated Testing/Validation, Packaging, Containerization

Machine Learning & AI: PyTorch, TensorFlow, Scikit-learn, XGBoost, Transformer Models (BERT), Recommendation Systems, Deep Learning, Neural Networks, MLOps, Computer Vision, Neural Reconstruction

Distributed Systems & Cloud: AWS (EC2, ECS, Lambda, Kinesis, EMR, SageMaker), Azure (AKS, Bicep), Kubernetes (EKS, AKS), Docker, Terraform, Apache Kafka, Apache Spark, Apache Flink, Kubebuilder, Istio, Cilium

Distributed Data & Storage: DynamoDB, PostgreSQL, Redis, MongoDB, Amazon Redshift, Amazon S3

Core CS: Data Structures & Algorithms, Distributed Systems (consensus, replication, sharding), Concurrency, System Design, Networking (TCP/IP, UDP, TLS, DNS, HTTP 3), Performance Optimization

Systems Programming: eBPF, Cilium, Lock-Free Data Structures, Memory Safety, Concurrency

EDUCATION

Bachelor of Technology — Computer Science & Engineering

RCC Institute of Information Technology — Kolkata, India | 2018